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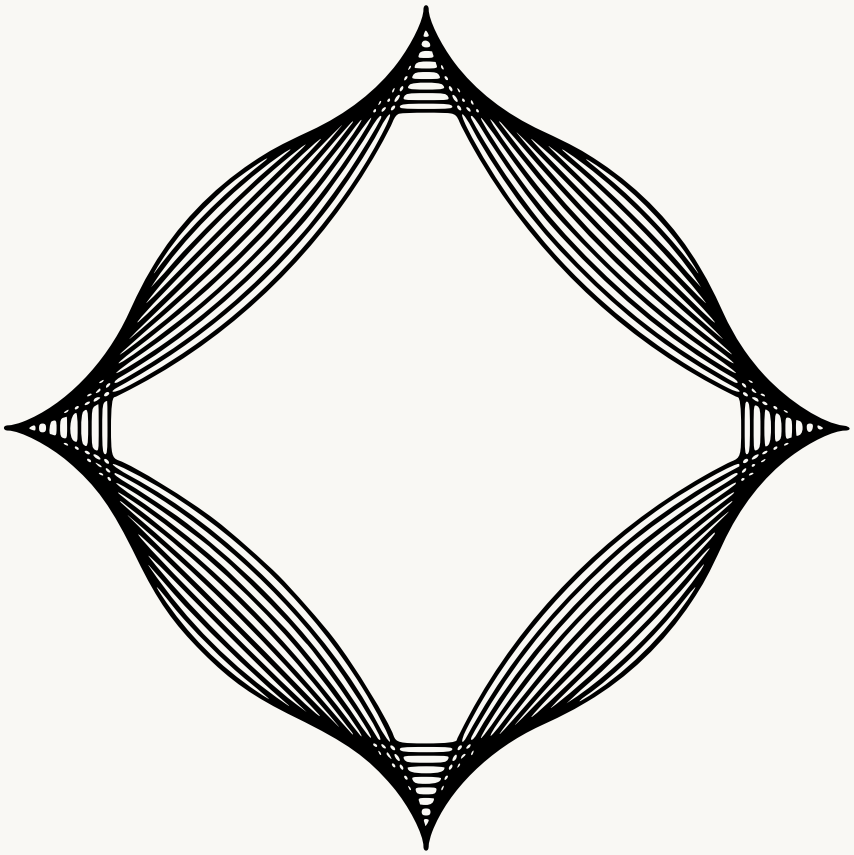

Quantum-Assisted Assurance for Autonomous Aviation Systems

A Research Concept for Training and Verifying Classical Autonomous Aircraft

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Motivation (The Problem)

FORMAL VERIFICATION IN THE AGE OF AUTONOMOUS AVIATION

- Autonomous aircraft now operate as distributed systems
- Failures arise from chains of decisions, not single actions
- Classical simulation cannot fully explore rare but dangerous cases
- Verification and certification are becoming the main bottleneck

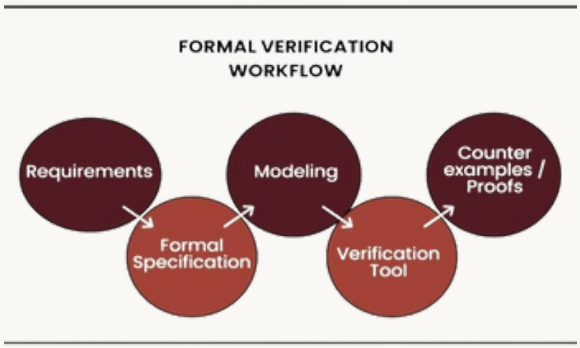
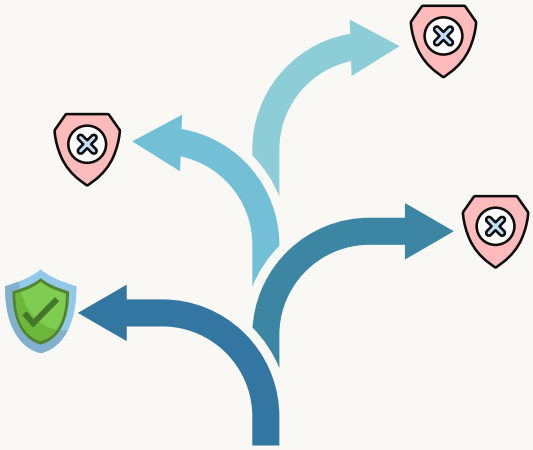


Fig 2: Formal Verification Workflow

Sunil Anasuri / IJERET, 3(1), 95-104, 2022

Formal verification provides strong guarantees when system behavior can be fully modeled. However, this assumption becomes increasingly difficult to maintain in autonomous aviation systems due to the following factors:

- Learning-based decision-making introduces non-static behavior
- State and action spaces grow exponentially with system complexity
- Safety-critical failures arise from long decision chains rather than single events



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What Is Not Being Proposed

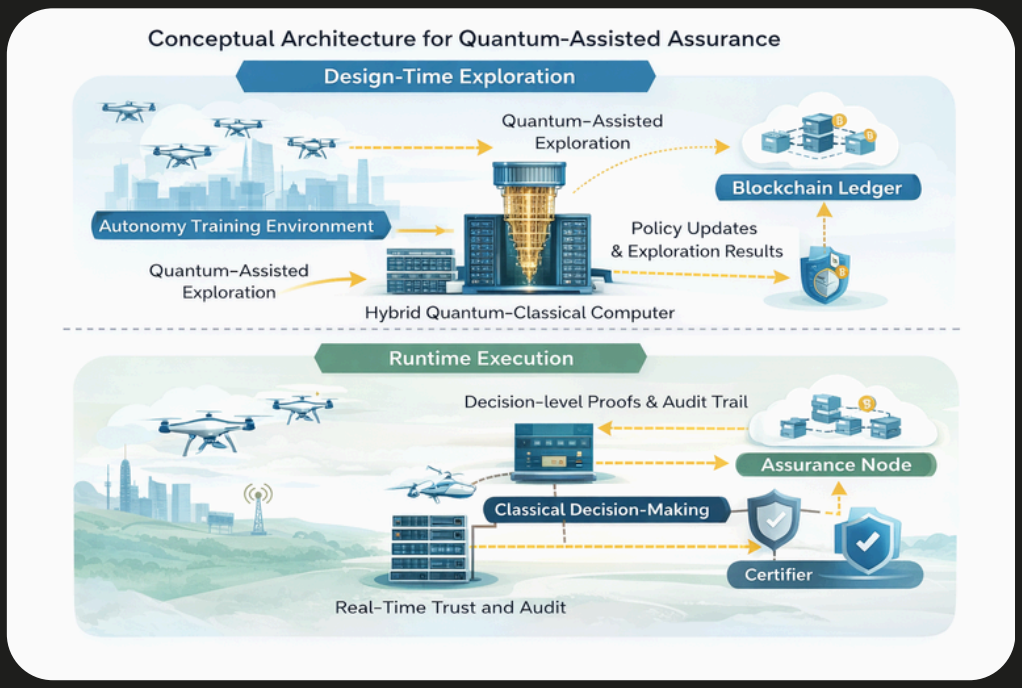
NOT THIS

- Quantum hardware onboard aircraft
- Quantum flight control
- Non-deterministic runtime behavior

THIS

- Quantum used off-board
- Assists training, testing, verification
- Aircraft runs fully classical AI

Quantum computation is treated as a design-time analytical tool, not a runtime decision-making component.



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Quantum-Assisted Assurance (QAA)

USE QUANTUM TO EXPLORE LARGE DECISION SPACES

Quantum-assisted methods are used off-board to efficiently explore complex autonomy decision spaces that are difficult to cover through classical simulation alone.

IDENTIFY RARE UNSAFE AUTONOMY BEHAVIORS

By searching low-probability regions of the decision space, quantum-assisted exploration helps reveal rare but safety-critical autonomy failures before deployment.

FEED INSIGHTS INTO CLASSICAL AI TRAINING

Findings from quantum-assisted analysis are fed into classical AI training to improve real time responses while preserving deterministic behavior.

DEPLOY ONLY CERTIFIABLE CLASSICAL AUTONOMY

All runtime autonomy remains fully classical, ensuring deterministic execution and compatibility with aviation certification standards.

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System Architecture

QUANTUM-ASSISTED DESIGN LAYER 01

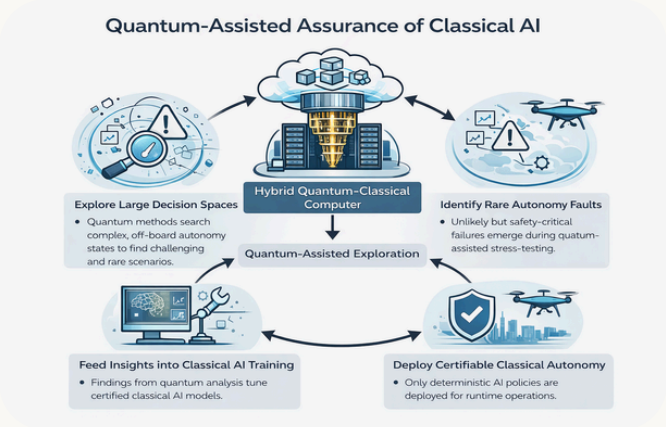
- Scenario exploration
- Policy stress-testing
- Off-board only

CLASSICAL AUTONOMOUS EXECUTION 02

- Deterministic AI
- Safety monitors
- Certifiable

ASSURANCE & VERIFICATION LAYER 03

- Decision summaries
- Explainability
- Audit support



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Research Questions

HOW CAN QUANTUM-GENERATED INSIGHTS BE DISTILLED INTO CLASSICAL, CERTIFIABLE AI?

CAN QUANTUM-ASSISTED EXPLORATION EXPOSE UNSAFE AUTONOMY BEHAVIORS FASTER THAN CLASSICAL SIMULATION?

WHAT DOES DECISION-LEVEL ASSURANCE MEAN FOR AUTONOMOUS AVIATION?

HOW CAN VERIFICATION SCALE WITHOUT AFFECTING REAL-TIME OPERATION?

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Methodology

PHASE 1: CLASSICAL BASELINE 01

- Establish decision policy
- Set safety metrics
- Define autonomy task

PHASE 2: QUANTUM-ASSISTED EXPLORATION 02

- Explore large decision spaces
- Surface rare unsafe behaviors

PHASE 3: HYBRID TRAINING 03

- Integrate quantum-derived insights
- Improve classical AI robustness
- Analyze decision traces

Role of ICDS Infra

Role of ICDS Infrastructure

High-Performance Simulation and Computation

ICDS high-performance computing resources enable large-scale autonomy simulations and repeated stress-testing across complex decision spaces that are infeasible on local systems.

Design-Time Exploration at Scale

The project uses off-board computation to explore rare and safety-critical autonomy behaviors, requiring parallelized workloads and high-throughput computational support aligned with ICDS capabilities.

Hybrid Quantum-Classical Experimentation

Quantum-assisted components are implemented using simulators and hybrid workflows that integrate classical computing with quantum-inspired or quantum-simulated optimization, all of which are well-suited to ICDS environments.

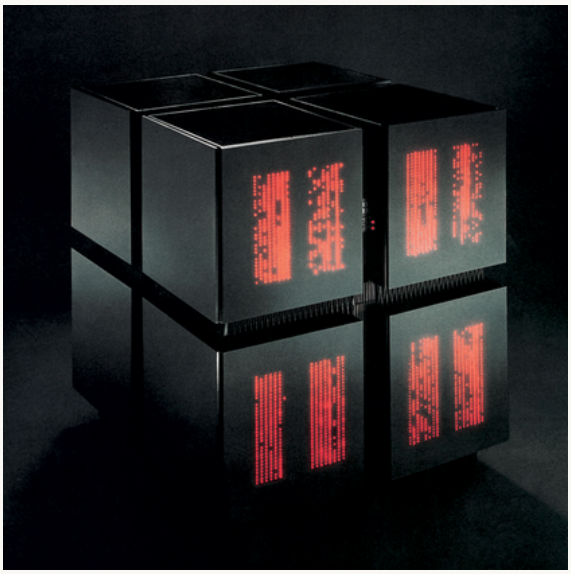

Photo: Steve Grohe, © Thinking Machines Corp

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